

Explaining Headline A/B Test Outcomes with Sparse Autoencoder Features

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Upworthy research archive

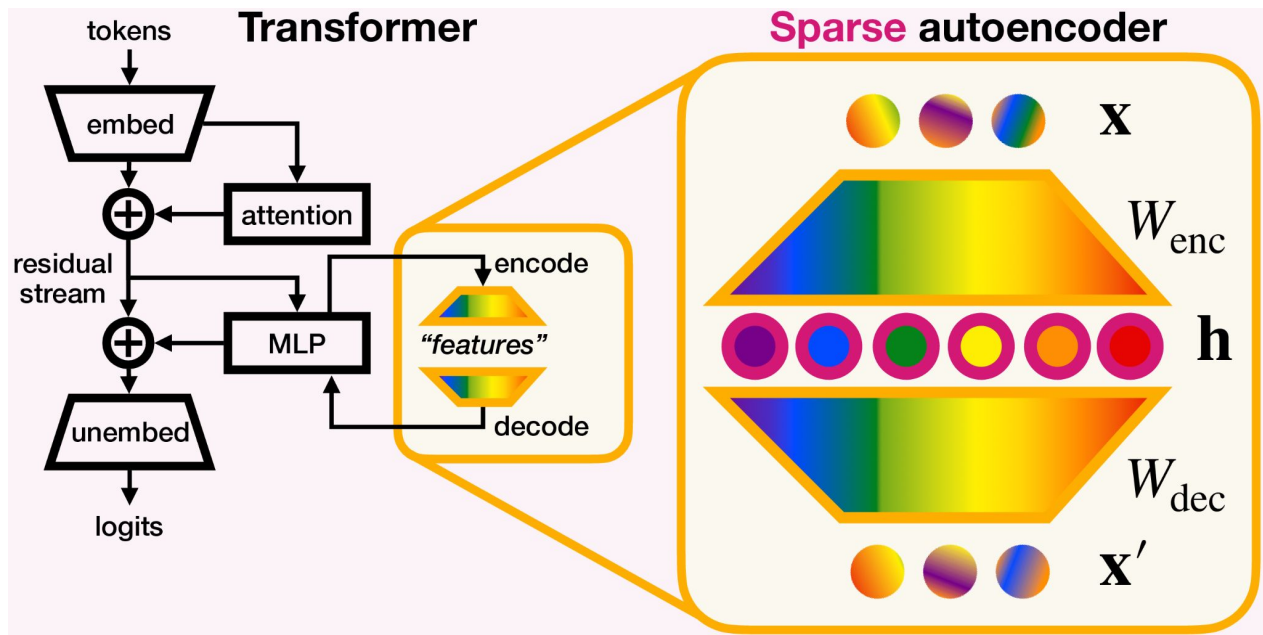
57k headlines

13k A/B tests

(average: 4.38)



Interpretable contextual embeddings

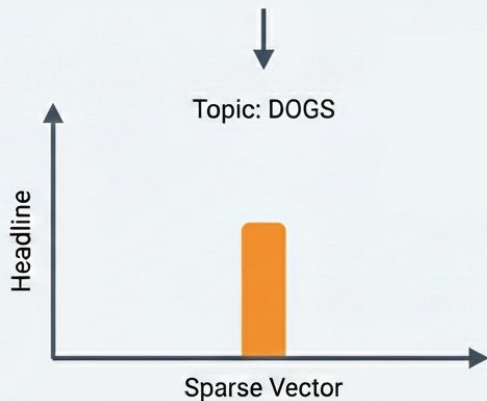


Gemma 3 - 22nd layer , GemmaScope 2, 16k features , Neuronpedia AutoInterp

Dataset construction

✗ WRONG: Single Headline Analysis

1 Simple Act That Saves The Lives Of
9000 Dogs ... Every Day

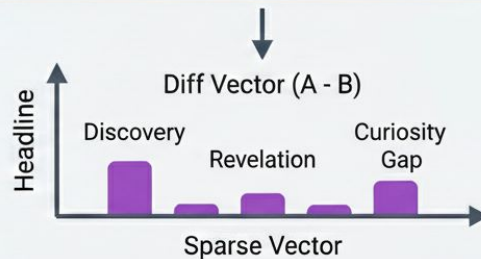


FLAW: Captures Topic Popularity, Not Writing Style.

✓ RIGHT: A/B Test Difference

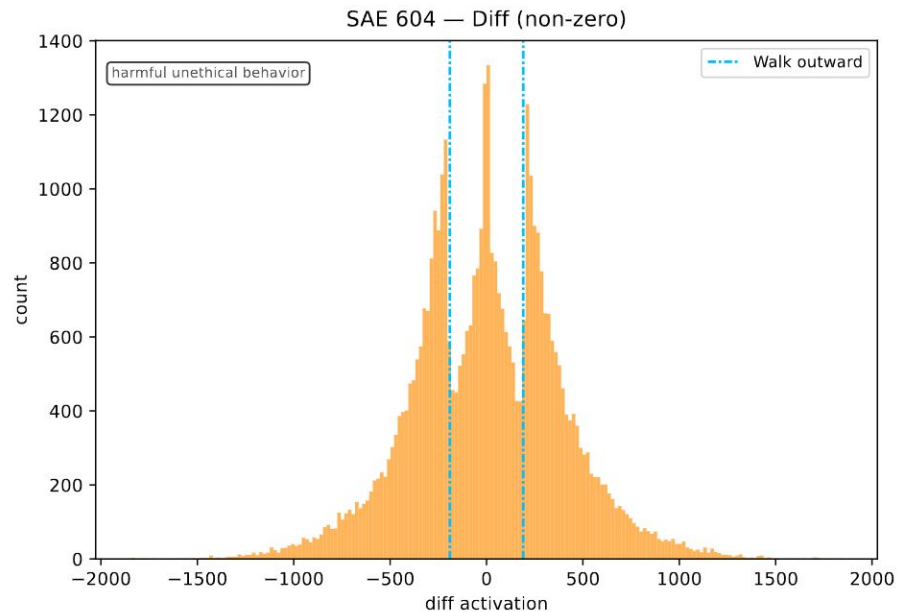
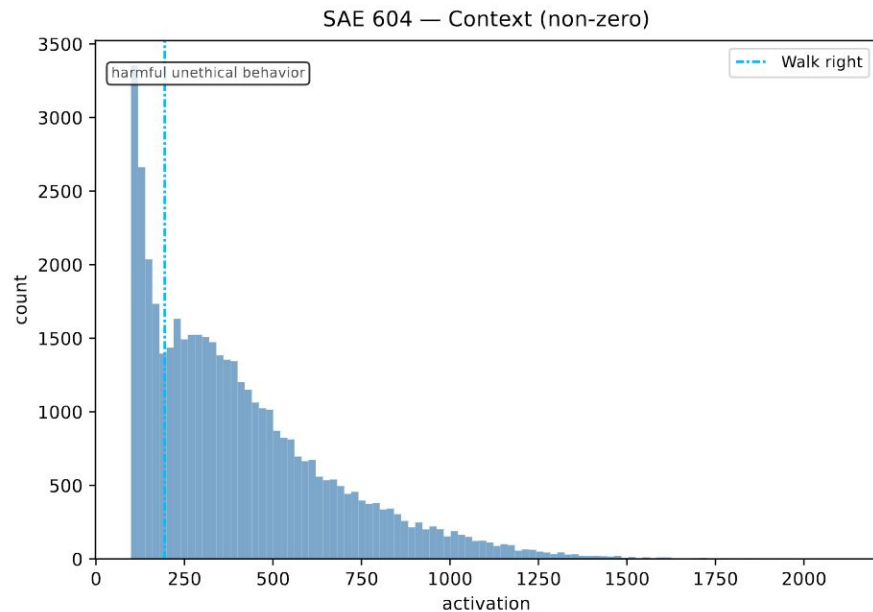
Winner (Headline A): "Dogs Would Tell Tell
Tell You If They Cuping You Could Talk"

Loser (Headline B): "1 Simple Act That
Saves The Lives Of 9000 Dogs ... Every Day"



BENEFIT: Isolates Specific Wording Choices.

Dataset construction



Interpretable machine learning

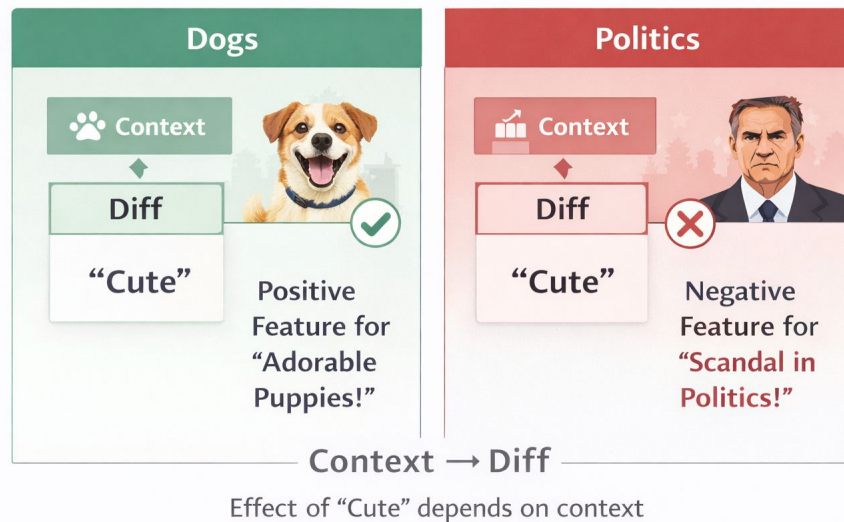
Logistic regression

Decision tree

XGBoost

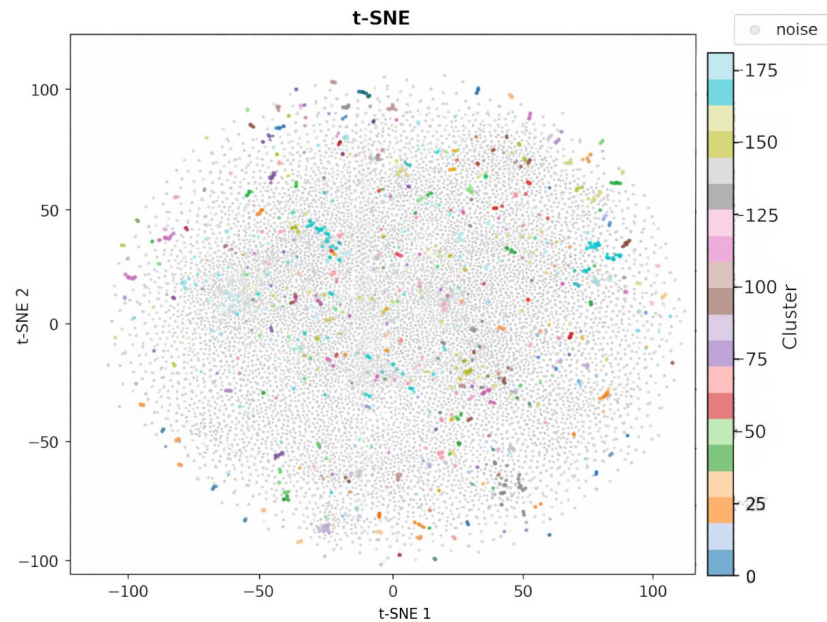
Decision rules

Hierarchy hypothesis



Hierarchical hypothesis

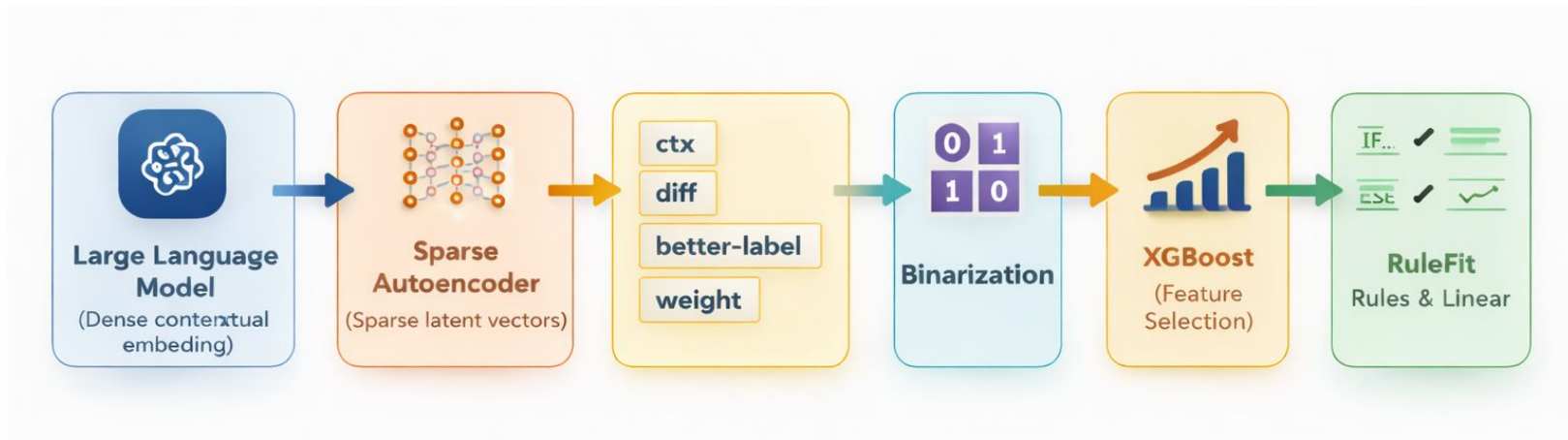
HDBSCAN Clusters on SVD-200D Context Vectors (182 clusters)



Results

#	Method	Acc	F1	AUC
1	Logistic Regression	0.606	0.606	0.750
2	Decision Tree	0.556	0.551	0.630
3	XGBoost (continuous)	0.616	0.616	0.759
4	XGBoost per-cluster	0.571	0.564	0.673
5	XGBoost (binarized)	0.618	0.619	0.759
6	XGBoost stumps ($d=1$)	0.577	0.578	0.681
7	Decision Tree (binarized)	0.553	0.549	0.625
8	RuleFit (467 features)	0.588	0.556	0.761

Final pipeline



Results

$\Delta f2220$	'this'	+0.1378	clickbait titles > 0.5	+0.07
$\Delta f11052$	How to write clickbait titles	+0.1078	unexpected things > 0.5	+0.06
$\Delta f1730$	remarkable and amazing qualities	+0.1130	intense negative emotions ≤ -0.5	-0.05
$\Delta f433$	problem, issue	+0.1020	recognizing or noticing ≤ -0.5	-0.14
$\Delta f9410$	subtitles	-0.1153		
$\Delta f1531$	discovery and revelation	+0.1168	joke, humor > -0.5 AND	
$\Delta f2223$	past actions or events	+0.1093	'thing' ≤ 0.5 AND	
$\Delta f1847$	shocking or horrifying scenarios	+0.1008	ctx: asking questions > 0.5	+0.03
$\Delta f604$	harmful unethical behavior	+0.0793		
$\Delta f1170$	quizzes and puzzles	+0.0916	regulation, risk > 0.5 AND	
$\Delta f2203$	stereotypes about gender roles	+0.0934	past events > -0.5	-0.04
$\Delta f11979$	sexual and explicit themes	+0.1482		
$\Delta f5439$	offensive content	+0.0907		
$\Delta f70$		-0.0742		
$\Delta f2161$	female names and women	+0.1025		

Results

Δf2220 “this”

- A: I Thought I Had This PSA Figured Out. Turns Out I Was Wrong. Big Time.
B: I Called Grandma After I Saw The Ending. You Probably Will, Too.
- A: I Thought I Had This PSA Figured Out. Turns Out I Was Wrong. Big Time.
B: An Eye-Opening Look Inside His Christmas Day Sheds Light On An Issue Too Many Of Us Ignore
- A: You Probably Didn't Learn This Sex History In School — But Holy Hot Peppers! Maybe You Should've.
B: The History Of Sex Is Interesting And Awkward — And You Should've Learned It.

Δf11052 “How to write clickbait titles”

- A: The Secret About Work Life In 20 Other Countries That Will Surprise You
B: In Japan, They Have A Word For Working Yourself To Death
- A: The Secret About Work Life In 20 Other Countries That Will Surprise You
B: The Next Time You Hear Someone Say That Americans Are Lazy, Show Them This
- A: When Everything Was Going Wrong For These Teens, A Kind Intervention Saved The Day
B: A Kid's Feelings Are Always Important, Especially When They're Not Heard

Δf1730 “remarkable and amazing qualities”

- A: Paul Walker's Unfinished Masterpiece Revealed. It's Not A Movie, It's Just Astonishingly Terrific.
B: How Come Paparazzis Never Manage To Catch Celebs Engaged In This Type Of Behavior?
- A: Photos Of An Astonishingly Beautiful Place That's Losing So Much Of What Makes It Beautiful
B: Photos Of Fading Beauty And The Story Of What's Taking It Away
- A: Photos Of Astonishingly Beautiful People Who Are Losing So Much Of What Makes Them Beautiful
B: Photos Of Fading Beauty And The Story Of What's Taking It Away

Results

Δf433 “problem, issue”

- A: There's One Huge Problem In Google And Facebook's Back Yard
B: You'll Be Shocked At What Exists In One Of The Richest Areas In The World
- A: There's One Huge Problem In Google And Facebook's Backyard
B: You'll Be Shocked At What Exists In One Of The Richest Areas In The World
- A: Wonder Woman, Spider-Man, Captain America. All These Heroes Are Awesome. There's Just One Problem.
B: Most Of Us Are Able To Find Comic Book Heroes That Look Like Us. How About The People Who Don't?

Δf9410 “subtitles”

- A: Where Have You Gone, FDR? A Nation Turns Its Broke-Ass Eyes To You.
B: The Stock Market Is Up, And I'd Prefer Not To Cheer...
- A: When The Price Is Wrong...So Wrong: America's Hospitals Get Wealthy This Way.
B: One Minute Of How Bad Our Healthcare System Really Is
- A: Get Used To Me: Black. Confident. Cocky.
B: These Kids Were Asked To Act Like A Famous Idol, And They Knocked It Out Of The Park

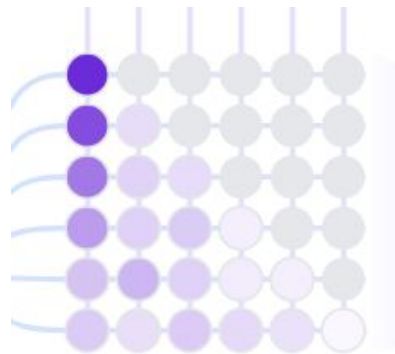
Δf1531 “discovery and revelation”

- A: On our first date, he deeply insulted me. I didn't realize until years later.
B: I'm a black guy who dates outside of my race. Here's a heart-dropping letdown moment I experienced.
- A: A creepy parody commercial for a pet company that made me so angry until I realized it was fake.
B: A couple gave their son a cute little puppy. So why am I sorta creeped out?
- A: I Thought This Ad Was Trying To Sell Me Jewelry, And Then I Realized It Had A Powerful Message
B: Marriage Gets A Shiny Reboot In This Fantastic Ad From India

Conclusion / Limitations

- long pipeline - accumulating error
- correlation vs causation
- crude clustering

Thank
you
for
your



Attention 🔍